

H4 - Adam Hand Trace

Problem Definition:

$$f(x, y) = 2x^2 + 6y^2$$

$$\epsilon = 10^{-8}$$

$$x^0 = (3, 2) \quad \alpha = 0.1, \beta_1 = 0.9, \beta_2 = 0.999$$

$$\text{Gradient: } \nabla f(x, y) = \begin{pmatrix} 4x \\ 12y \end{pmatrix}$$

Step-by-Step execution (4 decimals):

$$\text{Initialization: } m^0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, s^0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, k = 0$$

Iteration $k=0$:

$$x^0 = \begin{pmatrix} 3.0000 \\ 2.0000 \end{pmatrix} \quad g^0 = \begin{pmatrix} 12.0000 \\ 24.0000 \end{pmatrix}$$

$$m^1 = 0.9 \cdot m^0 + 0.1 \cdot g^0 = \begin{pmatrix} 1.2000 \\ 2.4000 \end{pmatrix}$$

$$s^1 = 0.999 \cdot s^0 + 0.001 \cdot (g^0)^2 = \begin{pmatrix} 0.1440 \\ 0.5760 \end{pmatrix}$$

$$\hat{m}^1 = m^1 / (1 - 0.9^1) = \begin{pmatrix} 12.0000 \\ 24.0000 \end{pmatrix}$$

$$\hat{s}^1 = \frac{s^1}{(1 - 0.999^1)} = \begin{pmatrix} 144.0000 \\ 576.0000 \end{pmatrix}$$

$$x^1 = x^0 - \alpha \frac{\hat{m}^1}{\sqrt{\hat{s}^1} + \epsilon} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} - 0.1 \begin{pmatrix} 12/12 \\ 24/24 \end{pmatrix} = \begin{pmatrix} 2.9000 \\ 1.9000 \end{pmatrix}$$

Iteration $k=1$:

$$x^1 = \begin{pmatrix} 2.9000 \\ 1.9000 \end{pmatrix} \quad g^1 = \begin{pmatrix} 11.6000 \\ 22.8000 \end{pmatrix}$$

$$m^2 = 0.9 \cdot m^1 + 0.1 \cdot g^1 = \begin{pmatrix} 1.0800 \\ 2.1600 \end{pmatrix} + \begin{pmatrix} 1.1600 \\ 2.2800 \end{pmatrix} = \begin{pmatrix} 2.2400 \\ 4.4400 \end{pmatrix}$$

$$s^2 = 0.999 \cdot s^1 + 0.001 \cdot (g^1)^2 = \begin{pmatrix} 0.2784 \\ 1.0953 \end{pmatrix}$$

$$\hat{m}^2 = m^2 / (1 - 0.9^2) = m^2 / 0.19 = \begin{pmatrix} 11.7895 \\ 23.3684 \end{pmatrix}$$

$$\hat{s}^2 = s^2 / (1 - 0.999^2) = \begin{pmatrix} 139.2776 \\ 547.9060 \end{pmatrix}$$

$$\sqrt{\hat{s}^2} = \begin{pmatrix} 11.8016 \\ 23.4074 \end{pmatrix}$$

$$x^2 = x^1 - 0.1 \cdot \begin{pmatrix} 11.7895/11.8016 \\ 23.3684/23.4074 \end{pmatrix} = \begin{pmatrix} 2.8001 \\ 1.8002 \end{pmatrix}$$

Table of Iterations

k	g	m	S	$\hat{m}$	$\hat{S}$	x^{k+1}
0	(12, 24)	(12, 2.4)	(0.144, 0.576)	(12, 24)	(144, 576)	(2.9000, 1.9000)
1	(11.6, 22.8)	(2.24, 10.2484, 1.0958, 4.44)	(0.2784, 1.0958)	(11.7895, 23.3684)	(139.2776, 547.9060)	(2.8001, 1.8002)

Comparison and Analysis.

A plain gradient step at $k=0$ with $\alpha = 0.1$ is $x^1 = x^0 - 0.1 \cdot (12, 24) \in (1.8, -0.4)$. The y -coordinate moves twice as far as the x -coordinate, overshooting heavily due to the steeper gradient. In contrast, Adam normalizes the step using the moving average of squared gradients ($\hat{S}$). Because the initial gradient is directly scaled by its own magnitude, Adam forces an exactly equal step size of 0.1 along both coordinate axes, dampening the dominant y -direction and accelerating the slower x -direction to ensure stable, diagonal descent.

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Signed: *Miras*